In Model III, $\mathrm{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given nonnegative definite matrix.

Consider next that H is possibly singular.

For any linear estimator $c + \underline{a}^{'}\underline{y}$ of an estimable function $\underline{\lambda}^{'}\underline{\beta}$, we have

$$\mathrm{Var}(c + \underline{a}^{'}\underline{y}) = \sigma^2 \underline{a}^{'} H \underline{a}.$$

Thus, the BLUE of an estimable function $\underline{\lambda}^{'}\underline{\beta}$ is given by $\underline{a}^{'}\underline{y}$ where $\underline{a}$ is any vector that minimizes $\underline{a}^{'} H \underline{a}$ subject to the constraints $X^{'}\underline{a} = \underline{\lambda}$.

The Lagrangian function for this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}^{'} H \underline{a} + 2\underline{\rho}^{'}(X^{'}\underline{a} - \underline{\lambda}).$$

We have

$$\frac{\partial L}{\partial \underline{a}} = 2H\underline{a} + 2X\underline{\rho} \quad \text{and} \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X^{'}\underline{a} - \underline{\lambda}).$$

Equating these partial derivatives to zero gives

$$H\underline{a} + X\underline{\rho} = \underline{0} \quad \text{and} \quad X^{'}\underline{a} = \underline{\lambda}.$$

Thus, a necessary condition for $\underline{a}^{'} H \underline{a}$ to assume a minimum at $\underline{a} = \widetilde{\underline{a}}$, for $\underline{a}$ satisfying $X^{'}\underline{a} = \underline{\lambda}$, is that the equations

$$\begin{bmatrix} H & X \\ X^{'} & 0 \end{bmatrix} \begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} \tag{1}$$

have a solution whose "first" component is $\widetilde{\underline{a}}$.

Take R to be any matrix such that $H = R'R$. [By Lemma 2.15, there exists an $r \times n$ matrix R such that $H = R'R$, where $r = \text{rank}(H)$.] Then, the above equations (1) can be written as

$$\begin{bmatrix} R'R & X \\ X' & 0 \end{bmatrix} \begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix},$$

which look very much like the earlier constrained normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

upon replacing

X with R (and $X'X$ with $H = R'R$)

A' with X

$\widehat{\underline{\beta}}$ with $\underline{a}$

$\underline{y}$ with $\underline{0}$

$\underline{b}$ with $\underline{\lambda}$

and $A\widehat{\underline{\beta}} = \underline{b}$ with $X'\underline{a} = \underline{\lambda}$.

Thus, if $\underline{\lambda}'\underline{\beta}$ is estimable, then by Theorem 2.12, the equations (1) are consistent.

Theorem 2.20. *Under Model III, a linear estimator $c + \underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE (that is, has minimum variance in the class of linear unbiased estimators) if and only if $c = 0$ and $\underline{a}$ is the first component of some solution to the equations (1).*

Proof. Take $\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix}$ to be any solution of equations (1) so $H\underline{a}+X\underline{\rho}=\underline{0}$ and $X^{'}\underline{a}=\underline{\lambda}$. Thus, $\underline{\lambda}^{'}=\underline{a}^{'}X$, implying that $\underline{a}^{'}\underline{y}$ is an unbiased estimator of $\underline{\lambda}^{'}\underline{\beta}$ (by Lemma 2.1). Take $k+\underline{b}^{'}\underline{y}$ as any other linear unbiased estimator of $\underline{\lambda}^{'}\underline{\beta}$ so $k=0$ and $X^{'}\underline{b}=\underline{\lambda}$.

Then,

$$\mathrm{Var}(k+\underline{b}^{'}\underline{y})=\mathrm{Var}(\underline{b}^{'}\underline{y})=\mathrm{Var}[\underline{a}^{'}\underline{y}+(\underline{b}-\underline{a})^{'}\underline{y}]=\mathrm{Var}(\underline{a}^{'}\underline{y})+\mathrm{Var}[(\underline{b}-\underline{a})^{'}\underline{y}]$$

$$=\mathrm{Var}(\underline{a}^{'}\underline{y})+(\underline{b}-\underline{a})^{'}(\sigma^2 H)(\underline{b}-\underline{a})=\mathrm{Var}(\underline{a}^{'}\underline{y})+\sigma^2(\underline{b}-\underline{a})^{'}R^{'}R(\underline{b}-\underline{a}).$$

Thus, $\mathrm{Var}(k+\underline{b}^{'}\underline{y})\geq\mathrm{Var}(c+\underline{a}^{'}\underline{y})$, with equality holding

Corollary 1. Under Model III, $c + \underline{a}^{'}\underline{y}$ is a BLUE of $\underline{\lambda}^{'}\underline{\beta}$ if and only if $H\underline{a} \in \mathcal{C}(X)$ and $E(c + \underline{a}^{'}\underline{y}) = \underline{\lambda}^{'}\underline{\beta}$.

Remark. Corollary 1 implies that $\underline{a}^{'}\underline{y}$ is a BLUE of its expectation if and only if $H\underline{a} \in \mathcal{C}(X)$.

Corollary 2. Under Model III, if $c + \underline{a}^{'}\underline{y}$ and $k + \underline{b}^{'}\underline{y}$ are both BLUEs of the same estimable function, then $\text{Var}[c + \underline{a}^{'}\underline{y} - (k + \underline{b}^{'}\underline{y})] = 0$ or equivalently $c + \underline{a}^{'}\underline{y} = k + \underline{b}^{'}\underline{y}$ with probability 1.

Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$.

For estimable $\underline{\lambda}'\underline{\beta}$ ($\underline{\lambda}' \in \mathcal{R}(X)$), a solution to equations (1) is given by

$$\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G_{12}\underline{\lambda} \\ G_{22}\underline{\lambda} \end{bmatrix}.$$

Because $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$ is symmetric, G' is also a generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$.

Another solution to the equations (1) is then given by

$$\begin{bmatrix} \underline{a} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}' \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G'_{11} & G'_{21} \\ G'_{12} & G'_{22} \end{bmatrix} \begin{bmatrix} \underline{0} \\ \underline{\lambda} \end{bmatrix} = \begin{bmatrix} G'_{21}\underline{\lambda} \\ G'_{22}\underline{\lambda} \end{bmatrix}.$$

Thus, under Model III, BLUEs of an estimable function $\underline{\lambda}'\underline{\beta}$ are given by $\underline{\lambda}'G'_{12}\underline{y}$ and $\underline{\lambda}'G_{21}\underline{y}$. By Corollary 2 to Theorem 2.20, $\underline{\lambda}'G'_{12}\underline{y} = \underline{\lambda}'G_{21}\underline{y}$ with probability 1.

Let $\Lambda'_1\underline{\beta}$ and $\Lambda'_2\underline{\beta}$ be estimable vectors. Then, the elements of $\Lambda'_1G'_{12}\underline{y}$ and $\Lambda'_2G'_{12}\underline{y}$ are BLUEs of the elements of $\Lambda'_1\underline{\beta}$ and $\Lambda'_2\underline{\beta}$, respectively.

If we replace $X'X$ with H and A' with X in Theorem 2.14 and Lemma 2.13, then these results hold for $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$, any generalized inverse of $\begin{bmatrix} H & X \\ X' & 0 \end{bmatrix}$, and can be used to show that,

$$\text{cov}[\Lambda'_1G'_{12}\underline{y}, \Lambda'_2G'_{12}\underline{y}] = \sigma^2\Lambda'_1(-G_{22})\Lambda_2,$$

and an unbiased estimator of σ^2 is given by

$$\widehat{\sigma}^2 = \underline{y}'G_{11}\underline{y}/\left[\text{rank}(H, X) - \text{rank}(X)\right].$$

Model IIIa, where q constraints $A\underline{\beta} = \underline{b}$ are imposed on the parameter space, can be approached as a special case of Model III.

The vector $\underline{b}$ can be regarded as a vector of q additional "data points," in which case the vector of all data points is given by $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$.

We have

$$\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix} = \begin{bmatrix} X \\ A \end{bmatrix} \underline{\beta} + \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix}$$

and

$$E \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \underline{0} \quad \text{and} \quad \text{Var} \begin{bmatrix} \underline{\epsilon} \\ \underline{0} \end{bmatrix} = \sigma^2 \begin{bmatrix} H & 0 \\ 0 & 0 \end{bmatrix}.$$

If we take the parameter space to be $\left\{\underline{\beta}, \sigma^2 : \underline{\beta} \in R^p, \sigma^2 > 0\right\}$, we arrive at an unconstrained model for $\begin{bmatrix} \underline{y} \\ \underline{b} \end{bmatrix}$ having the form of Model III.

It can be verified that BLUEs of any estimable function $\underline{\lambda}'\underline{\beta}$ computed on the basis of this model are BLUEs of $\underline{\lambda}'\underline{\beta}$ under the original constrained model, and vice versa.

2.11 When Is the Simple Least Squares Estimator a Best Linear Unbiased Estimator?

Take the model to be Model III, and suppose that $\underline{\lambda}'\underline{\beta}$ is an estimable function. The simple (or ordinary) least squares estimator (SLSE) of $\underline{\lambda}'\underline{\beta}$ is given by $\underline{\lambda}'\widetilde{\underline{\beta}}$, where $\widetilde{\underline{\beta}}$ is any solution to $X'X\widetilde{\underline{\beta}} = X'\underline{y}$. The SLSE is an unbiased estimator, but in general, it will not be a best linear unbiased estimator (BLUE).

Under what circumstances is the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ a BLUE? An answer is given by the following theorem.

Theorem 2.21. *Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that*

$$HX\underline{r} = X\underline{q},$$

where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.

Under what circumstances is the SLSE of every estimable function $\underline{\lambda}'\underline{\beta}$ a BLUE? An answer is provided by the following corollary.

Corollary 1. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a matrix Q such that

$$HX = XQ.$$

Proof. Note that $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X\underline{r} = \underline{\lambda}$ for some $\underline{r}$ (by Theorem 2.7). Thus, by Theorem 2.21, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if , for every $\underline{r}$, there exists a $\underline{q}$ such that

$$HX\underline{r} = X\underline{q}.$$

The next corollary gives a condition $HP_X = P_XH$ that is equivalent to $HX = XQ$ for some Q given in Corollary 1, where $P_X = X(X'X)^-X'$. Note that $(HP_X)' = P_X'H' = P_XH$ is not necessarily a symmetric matrix.

Corollary 2. Under Model III, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $HP_X = P_XH$ (that is, HP_X is a symmetric matrix).

Proof. By Corollary 1, it suffices to show that $HP_X = P_XH$ (that is, HP_X is a symmetric matrix) if and only if $HX = XQ$ for some Q.

Remarks. Theorem 2.21 and its two corollaries are still valid if $\text{Var}(\underline{y}) = \sigma^2 H$ is put in place of H.

Furthermore, under Model VII with variance V:

1. Theorem 2.21 and its two corollaries are still valid if V is put in place of H.

2. That is, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ if and only if there exists a vector $\underline{q}$ such that $VX\underline{r} = X\underline{q}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$;

3. And, under Model VII, the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$ for every $\underline{\lambda}$ for which $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists a Q such that $VX = XQ$ or if and only if $VP_X = P_X V$.

4. Above, elements of $\underline{q}$ and Q may be functions of $\underline{\theta}$.

5. For more details, see George Zyskind's paper, "On canonical forms, non-negative covariance matrices and best and simple least squares linear estimators in linear models" (*The Annals of Mathematical Statistics*, Volume 38, 1967, 1092–1109).

Example. Suppose that we have an observation on each of s variates for each of r "individuals." Let $\underline{y}_i = \begin{bmatrix} y_{i1} \\ \vdots \\ y_{ir} \end{bmatrix}$ represent the vector whose elements are the observations on the ith variate. Take the model equation for $\underline{y}_i$ to be

$$\underline{y}_i = U\underline{\beta}_i + \underline{\epsilon}_i,$$

where, for some t, U is a given $r \times t$ matrix and $\underline{\beta}_i$ is a $t \times 1$ vector of unknown parameters.

Suppose that

$$\text{Var} \begin{bmatrix} y_{1j} \\ \vdots \\ y_{sj} \end{bmatrix} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1s} \\ \vdots & \vdots & \vdots \\ \sigma_{s1} & \cdots & \sigma_{ss} \end{bmatrix},$$

where some or all of the $\sigma_{ii'}$'s may be unknown parameters; and suppose that $\text{cov}(y_{ij}, y_{i'j'}) = 0$ for $i = 1, \ldots, s$, $i' = 1, \ldots, s$, $j = 1, \ldots, r$, $j' = 1, \ldots, r$, and $j \neq j'$ (that is, observations from the same individuals may be correlated, but observations from different individuals are uncorrelated).

This model can be viewed as a special case of Model VII, where

$$\underline{y} = \begin{bmatrix} \underline{y}_1 \\ \vdots \\ \underline{y}_s \end{bmatrix}, \underline{\epsilon} = \begin{bmatrix} \underline{\epsilon}_1 \\ \vdots \\ \underline{\epsilon}_s \end{bmatrix}, \underline{\beta} = \begin{bmatrix} \underline{\beta}_1 \\ \vdots \\ \underline{\beta}_s \end{bmatrix}, X = \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix}, \text{ and } V = \begin{bmatrix} \sigma_{11}I_r & \cdots & \sigma_{1s}I_r \\ \vdots & \vdots & \vdots \\ \sigma_{s1}I_r & \cdots & \sigma_{ss}I_r \end{bmatrix},$$

and where the elements of $\underline{\theta}$ consist of the unknown $\sigma_{ii'}$'s.

Here, the $(i, i^{'})$th block of V is

$$V_{ii'} = \text{cov}(\underline{\epsilon}_i, \underline{\epsilon}_{i'}) = \text{cov}\left(\begin{bmatrix} \epsilon_{i1} \\ \vdots \\ \epsilon_{ir} \end{bmatrix}, \begin{bmatrix} \epsilon_{i'1} \\ \vdots \\ \epsilon_{i'r} \end{bmatrix} \right) = \begin{bmatrix} \text{cov}(\epsilon_{i1}, \epsilon_{i'1}) & \cdots & \text{cov}(\epsilon_{i1}, \epsilon_{i'r}) \\ \vdots & \vdots & \vdots \\ \text{cov}(\epsilon_{ir}, \epsilon_{i'1}) & \cdots & \text{cov}(\epsilon_{ir}, \epsilon_{i'r}) \end{bmatrix}$$

$$= \begin{bmatrix} \sigma_{ii'} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_{ii'} \end{bmatrix} = \sigma_{ii'} I_r.$$

Suppose that we wish to estimate an estimable function $\underline{\lambda}^{'}\underline{\beta} = \sum_{i=1}^{s} \underline{\lambda}_i^{'}\underline{\beta}_i$ [where $\underline{\lambda}^{'} = (\underline{\lambda}_1^{'}, \ldots, \underline{\lambda}_s^{'})$] of the elements of the parameter vector $\underline{\beta}$. Consider the estimator obtained by carrying out a separate least squares analysis on each variate, that is, the estimator $\sum_{i=1}^{s} \underline{\lambda}_i^{'}\widetilde{\underline{\beta}}_i$, where $U^{'}U\widetilde{\underline{\beta}}_i = U^{'}\underline{y}_i$.

Question. Is the estimator $\sum_{i=1}^{s} \underline{\lambda}_i^{'}\widetilde{\underline{\beta}}_i$ a BLUE of $\underline{\lambda}^{'}\underline{\beta}$ based on all the observations on all individuals?

Note that the normal equations $X^{'}X\widetilde{\underline{\beta}} = X^{'}\underline{y}$ based on all the observations on all individuals can be written as

$$\begin{bmatrix} U^{'}U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U^{'}U \end{bmatrix} \begin{bmatrix} \widetilde{\underline{\beta}}_1 \\ \vdots \\ \widetilde{\underline{\beta}}_s \end{bmatrix} = \begin{bmatrix} U^{'}\underline{y}_1 \\ \vdots \\ U^{'}\underline{y}_s \end{bmatrix}, \quad \text{or} \quad U^{'}U\widetilde{\underline{\beta}}_i = U^{'}\underline{y}_i,\ i = 1, \ldots, s.$$

Thus, the above question is equivalent to the question of whether the SLSE of an estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$. We have

$$VX = \begin{bmatrix} \sigma_{11}I_r & \cdots & \sigma_{1s}I_r \\ \vdots & \vdots & \vdots \\ \sigma_{s1}I_r & \cdots & \sigma_{ss}I_r \end{bmatrix} \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix} = \begin{bmatrix} \sigma_{11}U & \sigma_{12}U & \cdots & \sigma_{1s}U \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}U & \sigma_{s2}U & \cdots & \sigma_{ss}U \end{bmatrix}$$

$$= \begin{bmatrix} U & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & U \end{bmatrix} \begin{bmatrix} \sigma_{11}I_t & \sigma_{12}I_t & \cdots & \sigma_{1s}I_t \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}I_t & \sigma_{s2}I_t & \cdots & \sigma_{ss}I_t \end{bmatrix} = XQ$$

for

$$Q = \begin{bmatrix} \sigma_{11}I_t & \sigma_{12}I_t & \cdots & \sigma_{1s}I_t \\ \vdots & \vdots & \vdots & \vdots \\ \sigma_{s1}I_t & \sigma_{s2}I_t & \cdots & \sigma_{ss}I_t \end{bmatrix},$$

so that the SLSE of any estimable function $\underline{\lambda}'\underline{\beta}$ is a BLUE of $\underline{\lambda}'\underline{\beta}$.

2.12 Best Linear Unbiased Prediction

Suppose that the model for the data vector $\underline{y} = X\underline{\beta} + \underline{\epsilon}$ is Model III, with H nonsingular, and that we wish to predict (or estimate)

$$\underline{\lambda}^{'}\underline{\beta} + u,$$

where $u \in \mathbb{R}$ is a random variable having zero expectation $E(u) = 0$ and finite variance. For example, $\underline{\lambda}^{'}\underline{\beta} + u$ might correspond to a future data point.

A linear predictor (or estimator) $c + \underline{a}^{'}\underline{y}$ of $\underline{\lambda}^{'}\underline{\beta} + u$ is said to be unbiased if

$$E(c + \underline{a}^{'}\underline{y}) = \underline{\lambda}^{'}\underline{\beta} \text{ for all } \underline{\beta},$$

that is, if $E(c + \underline{a}^{'}\underline{y}) = E(\underline{\lambda}^{'}\underline{\beta} + u)$ for all $\underline{\beta}$. If there exists a linear unbiased predictor (or estimator) of $\underline{\lambda}^{'}\underline{\beta}+u$, then $\underline{\lambda}^{'}\underline{\beta}+u$ is said to be predictable (or estimable).

Suppose $\text{cov}(\underline{\epsilon}, u) = \sigma^2\underline{k}$, for some given vector $\underline{k}$, and $\underline{\lambda}^{'}\underline{\beta}$ is estimable.

One way to motivate predictors is to pretend (though this need not be true) that $(\underline{\epsilon}, u)$ are jointly normal: if so, the conditional expectation of u given $\underline{\epsilon}$ would be

$$E(u|\underline{\epsilon}) = \underline{k}^{'}H^{-1}\underline{\epsilon}.$$

In predicting $\underline{\lambda}^{'}\underline{\beta} + u$, we do not know $\underline{\beta}$ or $\underline{\epsilon}$, but we could estimate these.

Let $\widehat{\underline{\beta}} = (X^{'}H^{-1}X)^{-}X^{'}H^{-1}\underline{y}$ be a solution to $X^{'}H^{-1}X\widehat{\underline{\beta}} = X^{'}H^{-1}\underline{y}$.

Then, estimate $\underline{\lambda}^{'}\underline{\beta}$ by $\underline{\lambda}^{'}\widehat{\underline{\beta}}$ and predict u by $\widehat{u} \equiv \underline{k}^{'}H^{-1}\widehat{\underline{\epsilon}}$ using residuals $\widehat{\underline{\epsilon}} \equiv \underline{y} - X\widehat{\underline{\beta}}$.

That is, consider predicting $\underline{\lambda}^{'}\underline{\beta} + u$ by

$$\begin{aligned}
\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u} &\equiv \underline{\lambda}^{'}\widehat{\underline{\beta}} + \underline{k}^{'}H^{-1}\widehat{\underline{\epsilon}} \\
&\equiv \underline{\lambda}^{'}\widehat{\underline{\beta}} + \underline{k}^{'}H^{-1}(\underline{y} - X\widehat{\underline{\beta}}) \\
&= \underline{k}^{'}H^{-1}\underline{y} + (\underline{\lambda}^{'} - \underline{k}^{'}H^{-1}X)\widehat{\underline{\beta}} \\
&= \left[\underline{k}^{'}H^{-1} + (\underline{\lambda}^{'} - \underline{k}^{'}H^{-1}X)(X^{'}H^{-1}X)^{-}X^{'}H^{-1}\right]\underline{y} \equiv \underline{b}^{'}\underline{y},
\end{aligned}$$

where

$$\underline{b} \equiv H^{-1}\underline{k} + H^{-1}X\left[(X^{'}H^{-1}X)^{-}\right]^{'}(\underline{\lambda} - X^{'}H^{-1}\underline{k}).$$

By Theorem 2.19, we have

$$E(\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}) = E(\underline{\lambda}^{'}\widehat{\underline{\beta}}) + \underline{k}^{'}H^{-1}\left[E(\underline{y}) - E(X\widehat{\underline{\beta}})\right] = \underline{\lambda}^{'}\underline{\beta} + \underline{k}^{'}H^{-1}(X\underline{\beta} - X\underline{\beta}) = \underline{\lambda}^{'}\underline{\beta}.$$

Thus, $\underline{b}^{'}\underline{y} = \underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}$ is a linear unbiased predictor of $\underline{\lambda}^{'}\underline{\beta} + u$.

Let $c + \underline{a}^{'}\underline{y}$ be any linear unbiased predictor of $\underline{\lambda}^{'}\underline{\beta} + u$, requiring $\underline{a}^{'}X = \underline{\lambda}^{'}$ and $c = 0$.
One can show that mean squared error of $c + \underline{a}^{'}\underline{y}$ in prediction cannot be smaller than that of $\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}$, that is

$$E\left[c + \underline{a}^{'}\underline{y} - (\underline{\lambda}^{'}\underline{\beta} + u)\right]^2 \geq E\left[\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u} - (\underline{\lambda}^{'}\underline{\beta} + u)\right]^2.$$

Because $c + \underline{a}^{'}\underline{y}$ and $\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}$ are both linear unbiased predictors,

$$E\left[c + \underline{a}^{'}\underline{y} - (\underline{\lambda}^{'}\underline{\beta} + u)\right]^2 = \text{Var}\left[c + \underline{a}^{'}\underline{y} - (\underline{\lambda}^{'}\underline{\beta} + u)\right] = \text{Var}\left(\underline{a}^{'}\underline{y} - u\right)$$

and

$$E\left[\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u} - (\underline{\lambda}^{'}\underline{\beta} + u)\right]^2 = \text{Var}\left[\underline{b}^{'}\underline{y} - (\underline{\lambda}^{'}\underline{\beta} + u)\right] = \text{Var}\left(\underline{b}^{'}\underline{y} - u\right)$$

Write

$$\begin{aligned}
\text{Var}\left(\underline{a}^{'}\underline{y} - u\right) &= \text{Var}\left[(\underline{b}^{'}\underline{y} - u) + (\underline{a} - \underline{b})^{'}\underline{y}\right] \\
&= \text{Var}\left(\underline{b}^{'}\underline{y} - u\right) + \text{Var}\left[(\underline{a} - \underline{b})^{'}\underline{y}\right] + 2\text{cov}\left[(\underline{a} - \underline{b})^{'}\underline{y}, \underline{b}^{'}\underline{y} - u\right].
\end{aligned}$$

Therefore, among linear unbiased predictors of $\underline{\lambda}^{'}\underline{\beta} + u$, the predictor $\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}$ has uniformly smaller mean squared error than any other linear unbiased predictor. That is, $\underline{\lambda}^{'}\widehat{\underline{\beta}} + \widehat{u}$ is the unique best linear unbiased predictor (BLUP) of $\underline{\lambda}^{'}\underline{\beta} + u$.